# Artificial Intelligence Homework 3

April 16, 2026

**Q1: In our analysis of the wumpus world, we used the fact that each square contains a pit with probability $0.2$, independently of the contents of the other squares. Suppose instead that exactly $N/5$ pits are scattered at random among the $N$ squares other than $[1,1]$. Are the variables $P_{i,j}$ and $P_{k,l}$ still independent? What is the joint distribution $(P_{1,1},\ldots,P_{4,4})$ now? Redo the calculation for the probabilities of pits in $[1,3]$ and $[2,2]$ (conditional on *known*, $b$ as in section 13.6).**

**Q2: Consider two medical tests, A and B, for a virus. Test A is 95% effective at recognizing the virus when it is present, but has a 10% false positive rate (indicating that the virus is present, when it is not). Test B is 90% effective at recognizing the virus, but has a 5% false positive rate. The two tests use independent methods of identifying the virus. The virus is carried by 1% of all people. Say that a person is tested for the virus using only one of the tests, and that test comes back positive for carrying the virus. Which test returning positive is more indicative of someone really carrying the virus? Justify your answer mathematically.**

**Q3: The Markov blanket of a variable is defined on page 517. Prove that a variable is independent of all other variables in the network, given its Markov blanket and derive Equation (14.12)(page 538).**